Earth and Planetary Science Letters 288 (2009) 588–600

Contents lists available at ScienceDirect

Earth and Planetary Science Letters

journal homepage: www.elsevier.com/locate/epsl

The 1.35 Ga diabase sills from the northern North China Craton: Implications for breakup of the Columbia (Nuna) supercontinent

Shuan-Hong Zhang [a,b,*], Yue Zhao [a], Zhen-Yu Yang [a,c], Zhe-Feng He [a], Hai Wu [d]

[a] Institute of Geomechanics, Chinese Academy of Geological Sciences, Beijing 100081, China
[b] State Key Laboratory of Geological Processes and Mineral Resources, China University of Geosciences, Wuhan 430074, China
[c] Department of Earth Science, Nanjing University, Nanjing 210093, China
[d] Geological Samples Center, Ministry of Land and Resource, Sanhe 065201, China

ARTICLE INFO

Article history:
Received 6 August 2009
Received in revised form 15 October 2009
Accepted 15 October 2009
Available online 8 November 2009

Editor: R.W. Carlson

Keywords:
zircon
baddeleyite
mafic sill
U–Pb geochronology
continental breakup
Mid-Mesoproterozoic (Ectasian)
North China Craton (NCC)
Columbia (Nuna) supercontinent

ABSTRACT

Diabase sills are widespread within the Mesoproterozoic marine clastic and carbonate platformal sedimentary rocks in the northern North China Craton (NCC). Our new LA-ICP-MS zircon and baddeleyite dating on a diabase sill sample emplaced into the Mesoproterozoic Wumishan Formation yielded weighed mean $^{207}Pb/^{206}Pb$ ages of 1345 ± 12 Ma (95% confidence, MSWD = 1.8, N = 18) and 1353 ± 14 Ma (95% confidence, MSWD = 0.67, N = 15), respectively, indicating emplacement of the sills at the Mid-Mesoproterozoic (Ectasian) instead of the Late Paleozoic or Mesozoic as previously regarded. Existence of large volumes of 1.35 Ga diabase sills in the northern NCC indicates that the northern NCC underwent strong extension and mafic magmatism during Mid-Mesoproterozoic (Ectasian) time, probably related to the breakup of the Columbia (Nuna) supercontinent. Compared with the 1.4–1.3 Ga mafic dyke swarms (sills) in other cratons, we suggested that the NCC was a member of Nena including the Laurentia (North America and Greenland), Siberia and Baltica cratons in the supercontinent. The NCC was likely connected to the Siberia and North America (Laurentia) cratons during late Paleoproterozoic to Mid-Mesoproterozoic times, and breakup of the NCC away from the Columbia supercontinent occurred at around 1.35 Ga, which are consistent with the paleomagnetic results. The new zircon and baddeleyite geochronological results on the diabase sills provide not only magmatic evidence for Precambrian supercontinent reconstruction but also important constraint on ages of the Meso-Neoproterozoic strata in the NCC.

© 2009 Elsevier B.V. All rights reserved.

1. Introduction

Assembly, configuration, and breakup history and reconstruction of the paleo-supercontinents such as Columbia (Nuna), Rodinia and Pangea have been a topic of great interest to earth scientists during recent years (e.g., Dalziel, 1991; Hoffman, 1991; Moores, 1991; Windley, 1995; Karlstrom et al., 2001; Condie, 2002; Meert, 2002; Rogers and Santosh, 2002; Zhao et al., 2002; Rogers and Santosh, 2003; Zhao et al., 2003a; Kröner and Cordani, 2003; Zhao et al., 2004a, 2006; Ernst et al., 2008; Li et al., 2008; Murphy et al., 2009; Rogers and Santosh, 2009). Mafic dyke swarms (or sills), which are important indicators for widespread continental lithospheric extension, are very important for paleo-supercontinental reconstruction (e.g., Kamo et al., 1989; Harris and Li, 1995; Yale and Carpenter, 1998; Anderson and Cullers, 1999; Bleeker and Ernst, 2006; Holm et al., 2006; Ernst et al., 2008; Hou et al., 2008; Zhu et al., 2008).

The North China Craton (NCC) is regarded as a Precambrian craton that finally stabilized during the Paleoproterozoic around 1.85 Ga (e.g., Zhao et al., 2000, 2001; Wilde et al., 2002; Zhao et al., 2005). The basement of the NCC is composed of highly metamorphosed Archean and Paleoproterozoic rocks, which are covered by a thick sequence of unmetamorphosed Meso-Neoproterozoic (Changcheng, Jixian and Qingbaikou Systems) and Cambrian–Ordovician marine clastic and carbonate platformal sediments, Middle Carboniferous to Triassic fluvial and deltaic sediments, and Jurassic–Cretaceous and younger sediments and volcanic rocks (Figs. 1 and 2) (Li et al., 1995). After cratonization, the NCC underwent the late Paleoproterozoic extension with emplacement of mafic dyke swarms and anorthosite-mangerite-alkali granitoid-rapakivi granite suite in its northern and central parts in the period 1.78 to 1.68 Ga (e.g., Halls et al., 2000; Wang et al., 2004; Zhao et al., 2004b; Yang et al., 2005; Peng et al., 2005; Zhang et al., 2007a; Peng et al., 2008). Some K-rich volcanic rocks were indentified from the late Paleoproterozoic Dahongyu Formation with zircon U–Pb ages from 1622 ± 23 Ma to 1626 ± 9 Ma (Lu and Li, 1991; Lu et al., 2008; Gao et al., 2008a), which was likely related to continental rift process in the late Paleoproterozoic related to the dispersal of supercontinent Columbia (Lu et al., 2008). From the Mesoproterozoic

* Corresponding author. Address: No. 11 South Minzudaxue Road, Haidian District, Beijing 100081, China. Tel.: +86 10 68486987; fax: +86 10 68422326.
E-mail address: tozhangshuanhong@163.com (S.-H. Zhang).

0012-821X/$ – see front matter © 2009 Elsevier B.V. All rights reserved.
doi:10.1016/j.epsl.2009.10.023

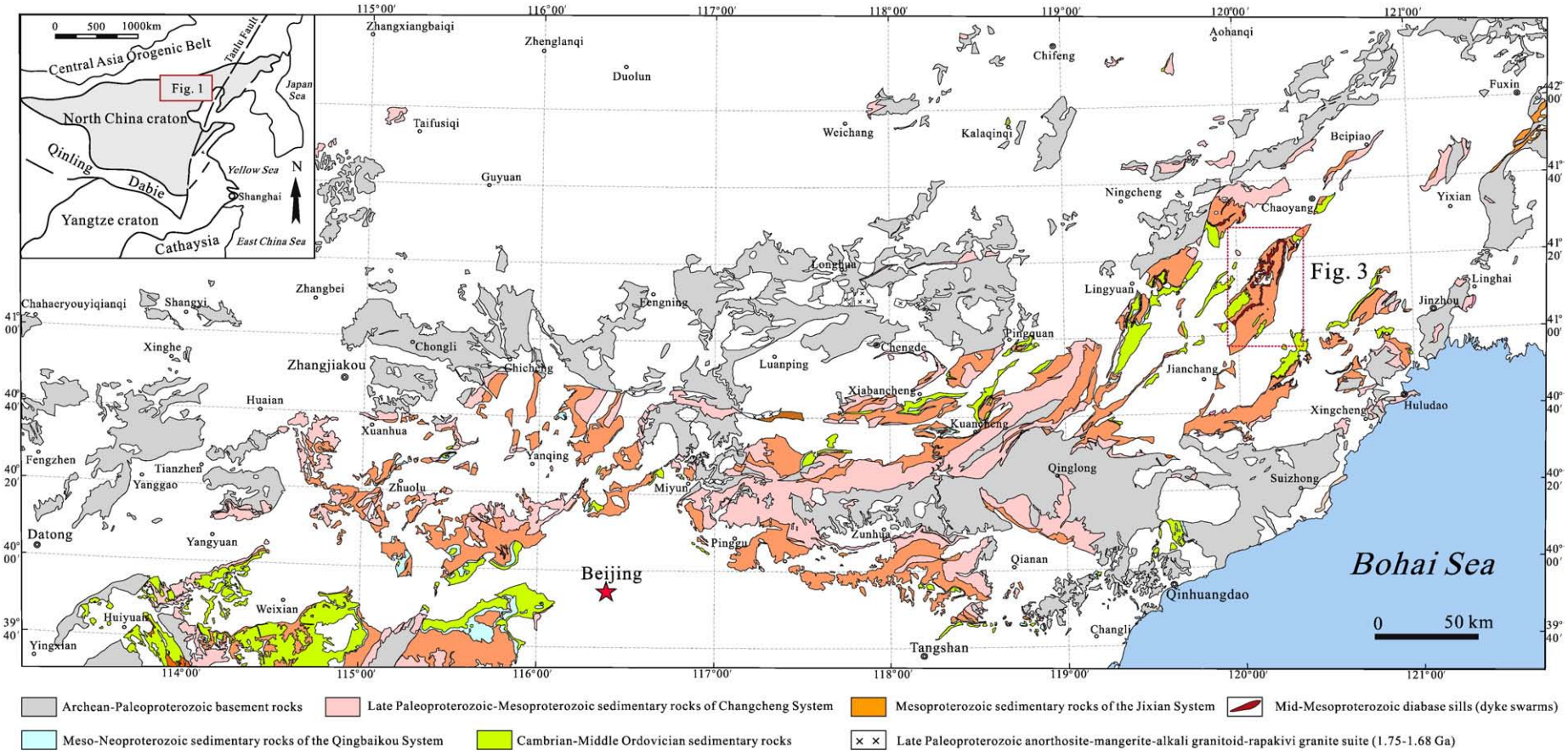


Fig. 1. Sketch of a geological map of the northern NCC showing distribution of the Archean–Paleoproterozoic basement rocks, the late Paleoproterozoic intrusive rocks and Meso-Neoproterozoic and Early Paleozoic sedimentary rocks.

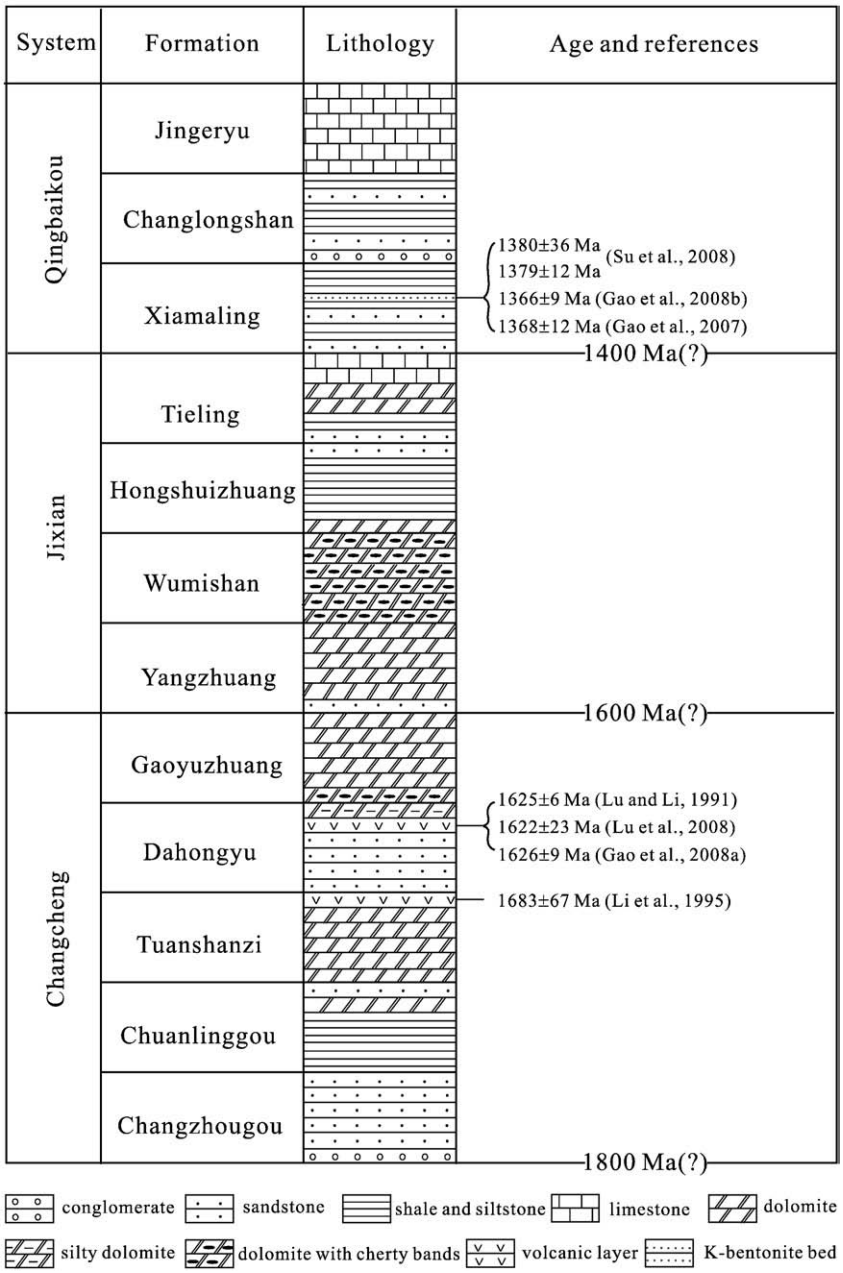


Fig. 2. Subdivision and zircon U–Pb age constraints on the Meso-Neoproterozoic strata in the northern NCC.

to Early Paleozoic, platformal carbonates are predominant in the NCC. Except for some Ordovician kimberlites in the eastern part, no Mesoproterozoic to Early Paleozoic magmatism occurred, indicating that the NCC was almost stable during this period.

Whether or how the NCC was involved in the Paleo-Mesoproterozoic supercontinents is of great interest to earth scientists in recent years (e.g., Zhai et al., 2000; Lu et al., 2002; Shao et al., 2002; Rogers and Santosh, 2002; Zhao et al., 2002, 2004a; Zhang et al., 2006; Lu et al., 2008; Li et al., 2008). Although some Mesoproterozoic granitoids and mafic dykes were reported in the northern NCC based on conventional single-grain zircon U–Pb ages (e.g., IMBGMR, 1997, 1999; HBGMR, 2003), recent zircon SHRIMP U–Pb dating results indicate that they are either Archean–Paleoproterozoic or Late Paleozoic in age (Zhang et al., 2007b, 2009a). Because of the lack of reliable ages for Meso-Neoproterozoic thermo-tectonic records in the NCC, the relationship between assembly, configuration, and breakup of the Proterozoic supercontinents and the NCC is still not clear.

Recently, some thin K-bentonite beds have been identified from the middle segment of the Xiamaling Formation in the northern NCC (Gao et al., 2007; Su et al., 2008). These K-bentonite beds are considered to be altered from felsic-intermediate tuffs and represent explosive volcanism in the northern NCC (Gao et al., 2007, 2008b; Su et al., 2008). SHRIMP U–Pb zircon analysis on four K-bentonite samples from different sites in the northern NCC yields similar weighed mean $^{207}Pb/^{206}Pb$ ages of 1368 ± 12 Ma, 1379 ± 12 Ma, 1366 ± 9 Ma and 1380 ± 36 Ma, respectively (Gao et al., 2007, 2008b; Su et al., 2008), indicating the formation of these tuffs during Mid-Mesoproterozoic (Ectasian) time. These ages provide important constraints on the ages of the Mesoproterozoic strata in the NCC. However, because volcanic tuffs, especially thin tuff beds can be transported for hundreds of kilometers in the air before deposition, the existence of Mid-Mesoproterozoic magmatism in the northern NCC is still uncertain. In this contribution, we report for the first time the zircon and baddeleyite geochronological results on a diabase sill sample within the Mesoproterozoic

sedimentary rocks, which provide important evidence for existence of Mid-Mesoproterozoic magmatism in the northern NCC and strong involvement of the NCC in the breakup of the Columbia supercontinent in the Mesoproterozoic.

2. Field occurrence and sample description

The diabase sills are widespread within the Mesoproerozoic marine clastic and carbonate platformal sedimentary rocks in the northern NCC and extend for several hundred kilometers from west to east and are especially common near Chaoyang, Lingyuan, Piangquan, Kuancheng and Xiabancheng (Fig. 1). They occur mainly within the Xiamaling and Wumishan Formations; locally within the Chuanlinggou, Gaoyuzhuang, Hongshuizhuang and Tieling Formations (Figs. 3 and 4). The Xiamaling Formation of the Qingbaikou System, which was previously regarded as Neoproterozoic in age (e.g., LBGMR, 1965; HBGMR, 1969; Lu et al., 2008), consists mainly of dark to black shales, sandstone, siltstone and marl layers. However, recent zircon SHRIMP U–Pb dating results of K-bentonite beds within the middle segment of the Xiamaling Formation indicate that the Xiamaling Formation was deposited during Mid-Mesoproterozoic (Ectasian) time. Diabase sills are very common within the Xiamaling Formation throughout the northern NCC. There are several layers of diabase sill with a thickness ranging from several meters to 100 m and sometimes exhibiting

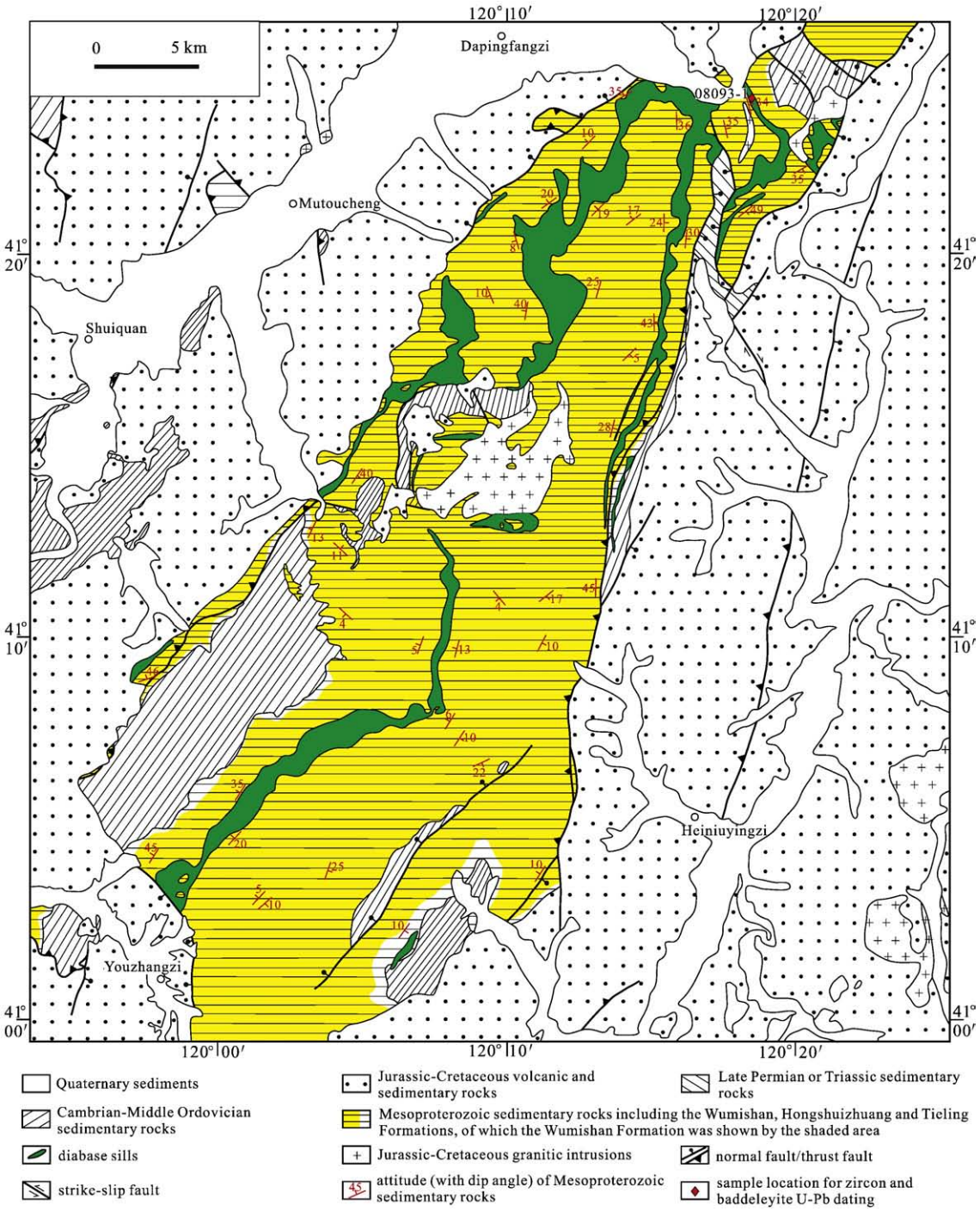


Fig. 3. Geological map showing distribution of the diabase sills within the Mesoproterozoic sedimentary rocks near Chaoyang in western Liaoning Province (modified after LBGMR, 1965, 1967, 1969).

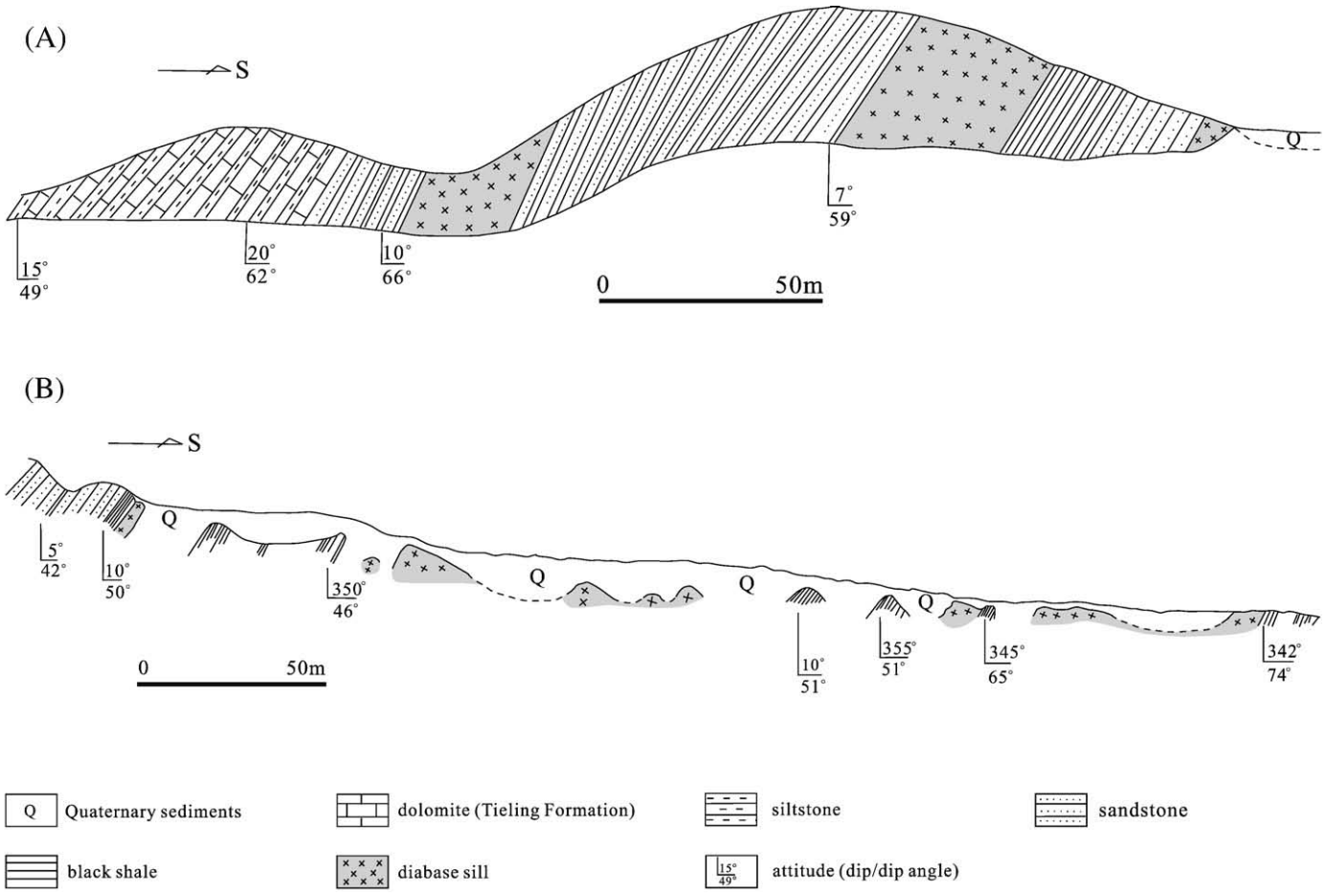


Fig. 4. Geological section showing the diabase sills within the Xiamaling Formation. (A) Wulongji section south to Xiabancheng; (B) Shuangdongzi section east to Pingquan.

concentric weathering (Fig. 5A). In Wulongji south to Xiabancheng, at least three diabase sills can be indentified (Fig. 4A), and in Shuangdongzi anticline east to Pingquan, four diabase sills can be indentified (Fig. 4B). Thin baked zones can be observed near the boundary of the diabase sills within the Xiamaling Formation. The Mesoproterozoic Wumishan Formation of the Jixian System is composed mainly of dolomitic limestone and dolomite with abundant cherty bands. Mafic sills are very common within the Wumishan Formation southwest to Chaoyang (Figs. 1 and 3). They are several ten meters to several hundred meters thick and kilometers to many tens of kilometers long. Although the boundary between the mafic sills and the dolomite (or dolomitic limestone) is irregular, the long axis of sills is almost parallel to the beding of strata, indicating their concordant emplacement along beds of host sedimentary rocks (Fig. 5D). Baked zones are very common and the dolomite (or dolomitic limestone) near the boundary displays recrystallization due to thermal emplacement of diabase sills (Fig. 5E). In some places, the boundaries between diabase sills and host rocks were reactivated as small faults (Fig. 5F). All the diabase rocks from the Xiamaling and Wumishan Formations display typical diabasic texture (Fig. 6) with similar mineral compositions of pyroxene (40–60 vol.%), plagioclase (40–55 vol.%), magnetite (3–10 vol.%) and hornblende (0–5 vol.%).

Although we tried to separate zircons and baddeleyites from over ten large diabase samples (>10 kg) emplaced within the Mesoproterozoic sedimentary rocks in different places in the northern NCC, only one sample (08093-1) from the sills emplaced into the Wumishan dolomitic limestone yields sufficient zircon and baddeleyite grains for dating. The sample is collected near a road cut (GPS position: 120°18.404′; 41°24.709′) southwest to Chaoyang (Fig. 2). It is composed of pyroxene (40 vol.%), plagioclase (55 vol.%) and magnetite (5 vol.%), with accessory apatite, zircon and baddeleyite. Zircons from this sample are semitransparent, creamy white subhedral long or short prisms with a size ranging from 60 to 150 μm and length/width ratio of 1:1 to 3.5:1 (Fig. 7A). Baddeleyites are semitransparent or opaque, brown euhedral platy prisms with a size ranging from 50 to 150 μm and length/width ratio of 1:1 to 6.5:1 (Fig. 7B).

3. Methods and analytical procedures

3.1. Dating method selection

Although it is not easy to separate zircons from the silica-unsaturated rocks such as diabase and gabbro for high precise in-situ U–Pb dating, it is still possible to obtain syn-magmatic zircons from these rocks and yield reliable ages (e.g., Kamo et al., 1989; Wingate et al., 1998; Holm et al., 2006; Zhu et al., 2008). Baddeleyite (ZrO_2), which usually exists in some mafic–ultramafic rocks, can provide another powerful tool to determine the emplacement ages (e.g., Kamo et al., 1989; Wingate et al., 1998; Söderlund and Johansson, 2002; Li et al., 2007; Heaman, 2009). Although baddeleyite $^{206}Pb/^{238}U$ ages could be affected by crystal orientation (Wingate and Compston, 2000), their $^{207}Pb/^{206}Pb$ ages are not affected. Therefore, baddeleyite $^{207}Pb/^{206}Pb$ ages can better reflect time of crystallization, especially for those older than 1.0 Ga. Moreover, baddeleyite has the advantage of being apparently less susceptible than zircon to the effects of Pb loss and alteration (Wingate and Compston, 2000). U–Pb baddeleyite dating in particular is a robust chronometer because it has high uranium contents, low common lead and is relatively rare in crustal rocks so it is relatively immune to crustal contamination (Heaman and LeCheminant, 1993; Holm et al., 2006). In contrast to

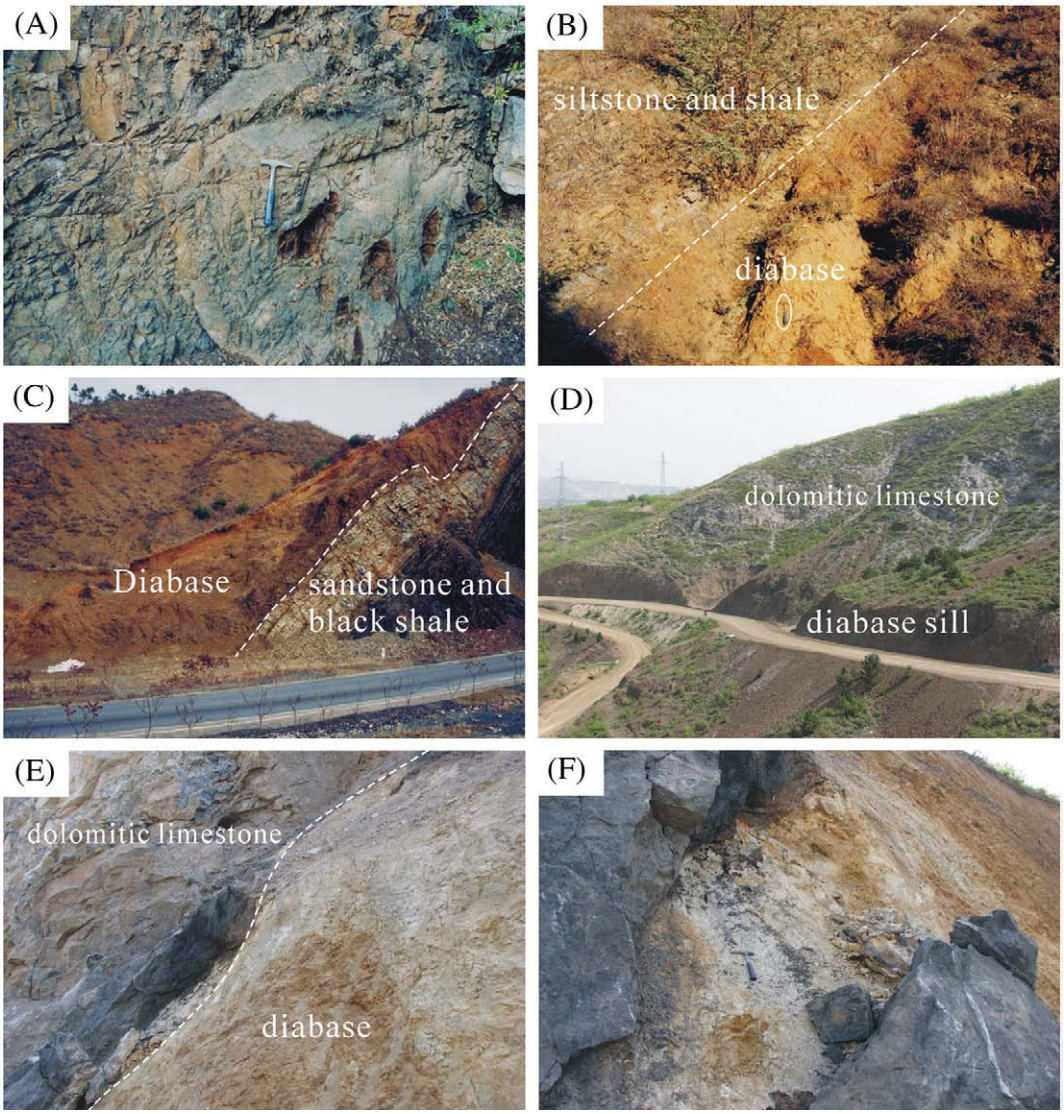


Fig. 5. Field photograph of typical diabase sills emplaced into the Mesoproterozoic sedimentary rocks. (A) concentric weathering of the diabase sills within the Xiamaling Formation near Shuangdongzi east to Pingquan, view to west; (B) diabase sills within the Xiamaling Formation near Shuangdongzi east to Pingquan, view to east; (C) diabase sills within the Xiamaling Formation near Wulongji south to Xiabancheng, view to east; (D) diabase sills emplaced along the bedding of the Wumishan Formation southwest to Chaoyang, view to north; (E) baked zones in boundary between diabase sill and Wumishan dolomitic limestone, view to north; (F) reactivated small faults and gouges near boundary between diabase sill and Wumishan dolomitic limestone, view to east.

zircons, baddeleyite inheritance in mafic intrusions is very unlikely (Krogh et al., 1987; Heaman and LeCheminant, 1993; Wingate, 2001), which makes baddeleyite more reliable for dating of the mafic rocks. Therefore, we dated both zircons and baddeleyite to determine the emplacement ages of the diabase sills.

3.2. Sample preparation and imaging

Zircons and baddeleyites were separated using conventional crushing and separation techniques and were then handpicked under a binocular microscope. They were mounted in epoxy resin and polished to expose the cores of the grains in readiness for photomicrograph, cathodoluminescence (CL), back-scattered electron (BSE) and LA-ICP-MS U–Pb isotopic analyses. Zircons were imaged using the HITACHI S3000-N electron microscope attached with GATAN Chroma CL detector at the Beijing SHRIMP Center; and baddeleyite BSE imaging was performed on a JEOL JXA-8800R microprobe at the Institute of Mineral Resources, Chinese Academy of Geological Sciences, Beijing. CL and BSE images of representative zircon and baddeleyite grains are shown in Fig. 7.

3.3. LA-ICP-MS U–Pb and trace element analysis

LA-ICP-MS U–Pb dating was performed on an excimer (193 nm wavelength) LA-ICP-MS at the State Key Laboratory of Geological Processes and Mineral Resources, China University of Geosciences, Wuhan and the State Key Laboratory of Continental Dynamics, Northwest University, Xi'an following the method described by Yuan et al. (2004) and Liu et al. (submitted for publication). The ICP-MS used is an Agilent 7500a. The GeoLas 2005 laser ablation system was used for the laser ablation experiments. Sites for dating were selected on the basis of CL and photomicrograph images. The spots used were 30–32 μm in diameter. Common Pb corrections were made using the method described by Andersen (2002). Because measured ^{204}Pb usually accounts for <0.35% of the total Pb (Appendix A), the correction is insignificant in most cases. U, Th, Pb and other trace element concentrations were calibrated by using ^{29}Si (for zircon) or ^{91}Zr (for baddeleyite) as an internal standard and GSE-1G as the reference standard. Isotopic ratios were calculated using ICPMS-DataCal 5.0 (Liu et al., submitted for publication), which were then corrected for both instrumental mass bias and depth-dependent

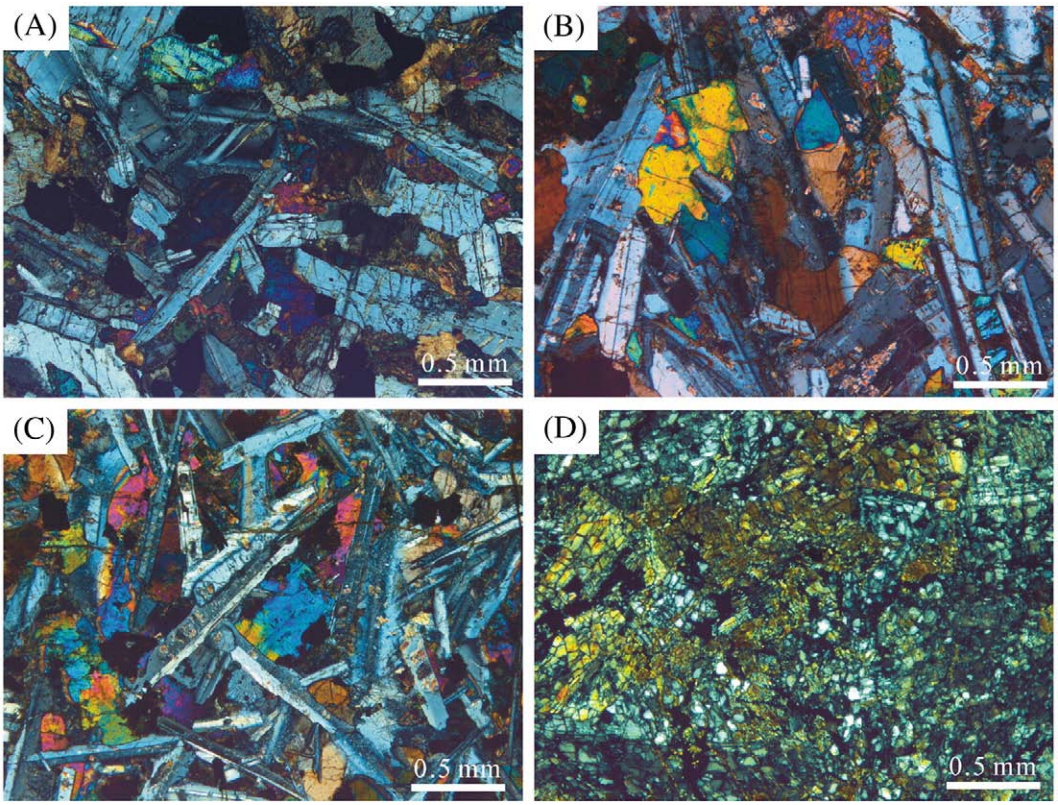


Fig. 6. Photomicrograph of typical diabase samples within the Mesoproterozoic sedimentary rocks. (A) Diabase emplaced into the Xiamaling Formation near Wulongji; (B–C) diabase emplaced into the Xiamaling Formation near Dayingzi northeast to Lingyuan; (D) altered diabase emplaced into the Wumishan Formation southwest to Chaoyang.

elemental and isotopic fractionation using Harvard zircon 91500 as external standard. Concordia diagrams and weighted mean ages were produced using the program ISOPLOT/Ex 3.23 (Ludwig, 2003).

4. Results

Twenty-five zircon grains from the Chaoyang diabase sample (08093-1) were analyzed using the LA-ICP-MS method. The analytical results are listed in Table 1 and plotted on a concordia diagram in Fig. 8A. Except for seven most anomalous spots (07, 09, 10, 15, 19, 22, and 23) caused by high common Pb contents, the remaining spots yield a weighted mean $^{207}Pb/^{206}Pb$ age of 1345 ± 12 Ma (95% confidence, MSWD = 1.8, N = 18; Fig. 8A), which is similar to the upper intercept age of the sample (1352 ± 15 Ma, MSWD = 0.45; the lower intercept age was forced through 0 Ma). High Th/U ratios (>1.55) and weak oscillatory zoning (Fig. 7A) suggest that they are magmatic zircons. Therefore, 1345 ± 12 Ma is interpreted as the crystallization age of the Chaoyang diabase sills.

Fifteen spots on fourteen baddeleyite grains from the same sample were also analyzed using the LA-ICP-MS method. The analytical results are listed in Table 1 and plotted on a concordia diagram in Fig. 8B. Although most of the analyses are not concordant (Fig. 8B), their $^{207}Pb/^{206}Pb$ ages are very similar. All the baddeleyite grains exhibit relatively homogenous concentrations of Th and U and show no or weak oscillatory zoning in BSE diagrams (Fig. 7B). The weighted mean $^{207}Pb/^{206}Pb$ age of the baddeleyite grains is 1353 ± 14 Ma (95% confidence, MSWD = 0.67, N = 15), which is similar to the upper intercept age of the sample (1359 ± 22 Ma, MSWD = 0.32; the lower intercept age was forced through 0 Ma). This weighted mean $^{207}Pb/^{206}Pb$ age is very similar to the zircon $^{207}Pb/^{206}Pb$ age obtained from the same sample, which reflects the crystallization age of the Chaoyang diabase sills.

5. Discussion

5.1. Emplacement age of the diabase sills within the Mesoproterozoic sedimentary rocks

Based on imprecise whole-rock K–Ar ages, the diabase sills emplaced into the Xiamaling Formation near Lingyuan and Kuancheng were previously regarded as Triassic in age (LBGMR, 1965; HBGMR, 1969), and those emplaced into the Wumishan Formation southwest to Chaoyang were considered as Late Paleozoic (Li and Li, 2005), Triassic (LBGMR, 1965, 1967) or Jurassic (LBGMR, 1969) in age. However, because of strong magmatic and tectonic activities during the Late Paleozoic to Mesozoic in the northern NCC (e.g., Zhao et al., 2004c; Zhang et al., 2009b), these K–Ar ages of diabase sills cannot be regarded as reliable ages. Therefore, more precise geochronological results are needed to determine the emplacement ages of these diabase sills.

Our new LA-ICP-MS dating on zircon and baddeleyite grains from a diabase sample emplaced into the Wumishan Formation yielded weighed mean $^{207}Pb/^{206}Pb$ ages of 1345 ± 12 Ma (95% confidence, MSWD = 1.8, N = 18) and 1353 ± 14 Ma (95% confidence, MSWD = 0.67, N = 15), respectively. The weighted mean $^{207}Pb/^{206}Pb$ age of the combined baddeleyite and zircon grains is 1348 ± 9 Ma (95% confidence, MSWD = 1.3, N = 33), which is almost similar to baddeleyite and zircon ages within errors. The agreement between baddeleyite and zircon dating results clearly indicates that the zircons in the diabase sample are primary (synmagmatic) and not inherited. Therefore, we concluded that the weighed mean $^{207}Pb/^{206}Pb$ ages of zircons and baddeleyites reflect the emplacement ages of the diabase sills within the Wumishan Formation. Considering the similar field occurrence and mineral compositions of the diabase sills within the Xiamaling and Wumishan Formations, we inferred that the large volumes of diabase sills within the Mesoproterozoic sedimentary rocks were emplaced at around 1.35 Ga.

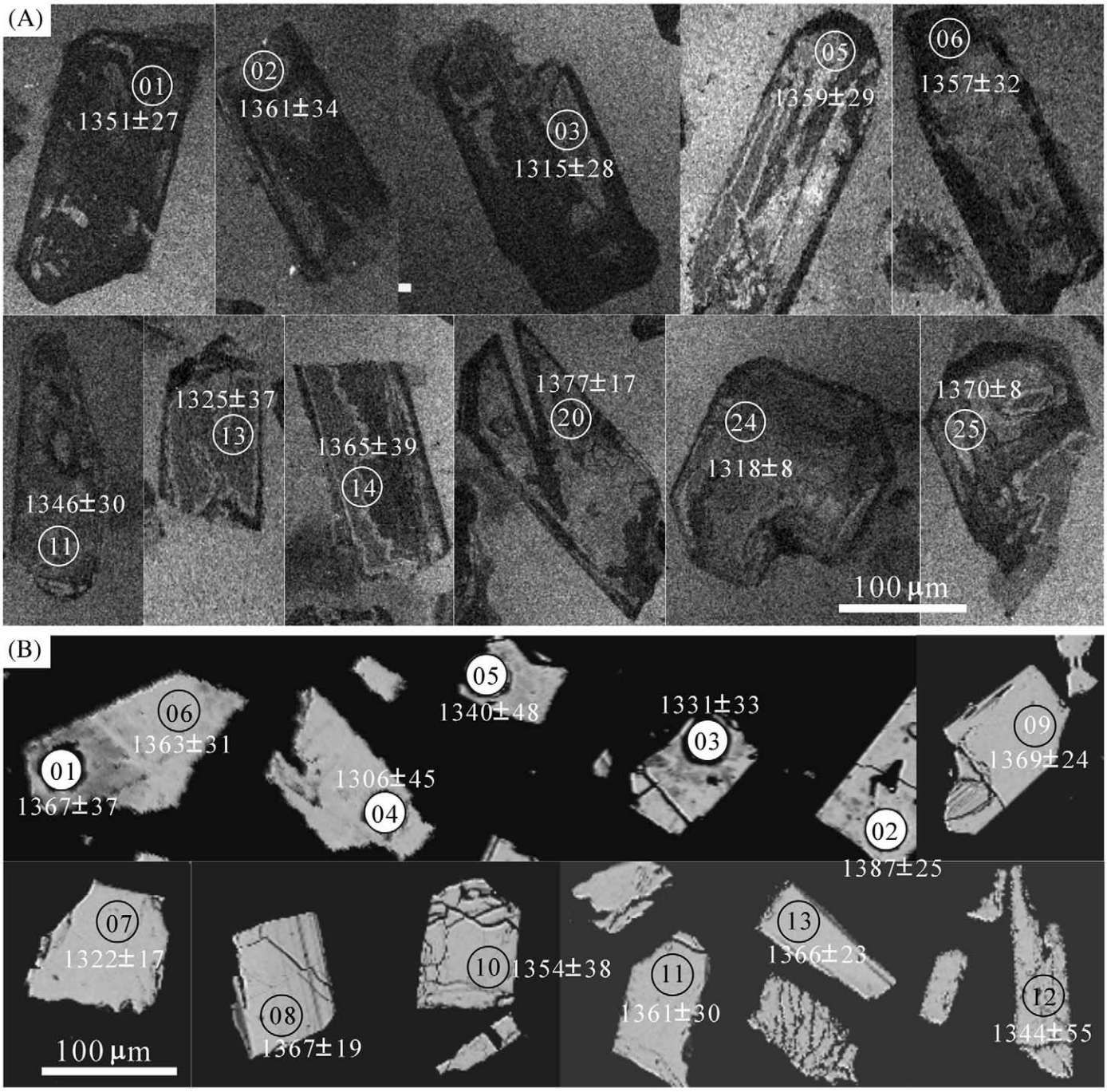


Fig. 7. Representative CL images of zircon grains (A) and BSE images of baddeleyite (B) grains, together with LA-ICP-MS U–Pb analysis spots and $^{207}Pb/^{206}Pb$ ages (Ma).

5.2. Tectonic implications

Paleo-Mesoproterozoic (1.8–1.2 Ga) is regarded as a period for assembly, growth and breakup of the Columbia (Nuna) supercontinent (Zhao et al., 2002, 2003a, 2004a; Ernst et al., 2008). The NCC was formed by at least two Paleoproterozoic collisional events, with the earlier one forming the east–west-trending Khondalite Belt along which the Yinshan Block in the north and the Ordos Block in the south amalgamated to form the Western Block, which then collided with the Eastern Block along the Trans-North China Orogen to form the coherent basement of the NCC at about 1.85 Ga (Zhao et al., 2000, 2001, 2005; Kröner et al., 2006; Xia et al., 2006). Following its final assembly at about 1.85 Ga, the NCC probably experienced subduction-related outgrowth along its southern margin, forming the Xiong'er volcanic belt (e.g., He et al., 2008; Zhao et al., 2009). These Paleoproterozoic collisional events and Paleo-Mesoproterozoic accretionary events recorded in the NCC have been considered as evidence for the assembly and outgrowth of the NCC within the Columbia supercontinent (Zhao et al., 2002, 2004a; He et al., 2008; Zhao et al., 2009).

Because of the lack of reliable isotopic ages for the Mesoproterozoic (1.6–1.0 Ga) magmatism and tectonism in the NCC, whether or how the NCC was involved in the assembly and breakup of the Paleo-Mesoproterozoic supercontinents (Columbia and Rodinia) and the position of the NCC in this supercontinents still remain unclear or controversial. Zhao et al. (2003a, 2004a) suggested that the Zhaertai-

Table 1
LA-ICP-MS U–Pb dating results of zircons and baddeleyites from the diabase sills in the northern NCC.

Grain-spot	Pb (ppm)	^{232}Th (ppm)	^{238}U (ppm)	^{232}Th/^{238}U	^{204}Pb/^{206}Pb	^{207}Pb^{a}/^{206}Pb^{a}	^{207}Pb^{a}/^{235}U	^{206}Pb^{a}/^{238}U	^{207}Pb^{a}/^{206}Pb^{a} Age (Ma)	^{207}Pb^{a}/^{235}U Age (Ma)	^{206}Pb^{a}/^{238}U Age (Ma)
Zircons from sample 08093-1											
01	881	9444	3465	2.73	0.00070	0.0866 ± 0.0012	1.810 ± 0.058	0.1505 ± 0.0043	1351 ± 27	1049 ± 21	904 ± 24
02	104	1013	652	1.55	0.00049	0.0870 ± 0.0016	1.390 ± 0.069	0.1174 ± 0.0059	1361 ± 34	885 ± 29	716 ± 34
03	377	2498	1062	2.35	0.00010	0.0850 ± 0.0012	2.588 ± 0.039	0.2199 ± 0.0018	1315 ± 28	1297 ± 11	1281 ± 10
04	304	1896	1206	1.57	0.00014	0.0837 ± 0.0013	2.115 ± 0.040	0.1820 ± 0.0020	1287 ± 30	1154 ± 13	1078 ± 11
05	700	6847	2062	3.32	0.00019	0.0869 ± 0.0013	2.137 ± 0.036	0.1776 ± 0.0017	1359 ± 29	1161 ± 12	1054 ± 9
06	708	6945	2114	3.28	0.00030	0.0868 ± 0.0015	2.069 ± 0.054	0.1718 ± 0.0035	1357 ± 32	1139 ± 18	1022 ± 19
07	1050	11651	4114	2.83	0.00126	0.0983 ± 0.0016	1.853 ± 0.042	0.1363 ± 0.0023	1592 ± 36	1065 ± 15	824 ± 13
08	1160	12195	2609	4.68	0.00014	0.0866 ± 0.0012	2.339 ± 0.039	0.1954 ± 0.0022	1352 ± 32	1224 ± 12	1150 ± 12
09	283	3097	1358	2.28	0.00120	0.0953 ± 0.0017	1.707 ± 0.108	0.1263 ± 0.0064	1544 ± 33	1011 ± 41	767 ± 37
10	609	6392	2910	2.20	0.00174	0.0995 ± 0.0014	1.767 ± 0.050	0.1282 ± 0.0032	1617 ± 26	1033 ± 18	778 ± 18
11	886	7590	2802	2.71	0.00014	0.0863 ± 0.0013	2.170 ± 0.038	0.1815 ± 0.0017	1346 ± 30	1172 ± 12	1075 ± 9
12	823	6786	2764	2.45	0.00021	0.0861 ± 0.0014	2.154 ± 0.072	0.1794 ± 0.0047	1340 ± 31	1166 ± 23	1064 ± 26
13	381	1863	1146	1.63	0.00010	0.0854 ± 0.0014	2.747 ± 0.047	0.2324 ± 0.0017	1325 ± 37	1341 ± 13	1347 ± 9
14	138	783	410	1.91	0.00028	0.0872 ± 0.0018	2.693 ± 0.060	0.2235 ± 0.0027	1365 ± 39	1327 ± 16	1300 ± 14
15	803	16078	2457	6.54	0.00166	0.0976 ± 0.0014	1.772 ± 0.036	0.1308 ± 0.0018	1589 ± 26	1035 ± 13	793 ± 10
16	167	1534	605	2.54	0.00000	0.0863 ± 0.0013	2.055 ± 0.066	0.1715 ± 0.0048	1346 ± 28	1134 ± 22	1020 ± 27
17	149	727	452	1.61	0.00010	0.0874 ± 0.0013	2.783 ± 0.044	0.2300 ± 0.0017	1370 ± 28	1351 ± 12	1334 ± 9
18	260	1350	768	1.76	0.00000	0.0864 ± 0.0012	2.731 ± 0.040	0.2286 ± 0.0016	1346 ± 27	1337 ± 11	1327 ± 8
19	946	6562	2712	2.42	0.00174	0.1067 ± 0.0019	2.878 ± 0.081	0.1941 ± 0.0043	1744 ± 27	1376 ± 21	1143 ± 23
20	441	3806	1670	2.28	0.00126	0.0877 ± 0.0008	1.737 ± 0.082	0.1420 ± 0.0030	1377 ± 17	1022 ± 31	856 ± 17
21	675	3510	1836	1.91	0.00029	0.0859 ± 0.0006	2.410 ± 0.100	0.2010 ± 0.0022	1339 ± 13	1246 ± 30	1181 ± 12
22	716	4436	1736	2.56	0.00176	0.1047 ± 0.0010	2.588 ± 0.109	0.1788 ± 0.0034	1709 ± 17	1297 ± 31	1060 ± 18
23	689	4406	2807	1.57	0.00156	0.0991 ± 0.0006	1.880 ± 0.077	0.1367 ± 0.0017	1607 ± 12	1074 ± 27	826 ± 10
24	1512	9032	4314	2.09	0.00025	0.0851 ± 0.0004	2.136 ± 0.097	0.1813 ± 0.0033	1318 ± 8	1161 ± 32	1074 ± 18
25	583	5457	1067	5.11	0.00087	0.0874 ± 0.0006	2.281 ± 0.110	0.1891 ± 0.0021	1370 ± 8	1206 ± 34	1117 ± 11
Baddeleyites from sample 08093-1											
01	83	5	228	0.02	0.00017	0.0868 ± 0.0017	4.679 ± 0.175	0.3855 ± 0.0109	1367 ± 37	1764 ± 31	2102 ± 51
02	175	28.9	599	0.05	0.00000	0.0882 ± 0.0014	3.691 ± 0.104	0.3028 ± 0.0076	1387 ± 25	1569 ± 22	1705 ± 38
03	84	14.6	269	0.05	0.00016	0.0857 ± 0.0014	4.063 ± 0.130	0.3409 ± 0.0089	1331 ± 33	1647 ± 26	1891 ± 43
04	72	2.79	185	0.02	0.00000	0.0846 ± 0.0015	5.041 ± 0.136	0.4299 ± 0.0086	1306 ± 35	1826 ± 23	2305 ± 39
05	60	18.9	184	0.10	0.00042	0.0861 ± 0.0017	3.950 ± 0.092	0.3320 ± 0.0049	1340 ± 38	1624 ± 19	1848 ± 24
06	59	4	190	0.02	0.00030	0.0871 ± 0.0014	3.125 ± 0.145	0.2596 ± 0.0045	1363 ± 31	1439 ± 36	1488 ± 23
07	165	13	414	0.03	0.00009	0.0853 ± 0.0007	4.005 ± 0.143	0.3398 ± 0.0027	1322 ± 17	1635 ± 29	1886 ± 13
08	98	15	261	0.06	0.00011	0.0868 ± 0.0008	3.949 ± 0.143	0.3292 ± 0.0054	1367 ± 19	1624 ± 29	1834 ± 26
09	86	15	231	0.06	0.00022	0.0873 ± 0.0011	3.735 ± 0.150	0.3074 ± 0.0060	1369 ± 24	1579 ± 32	1728 ± 29
10	149	45	440	0.10	0.00012	0.0867 ± 0.0017	3.356 ± 0.150	0.2784 ± 0.0052	1354 ± 38	1494 ± 35	1583 ± 26
11	69	9	201	0.05	0.00014	0.0870 ± 0.0013	3.550 ± 0.134	0.2932 ± 0.0052	1361 ± 30	1538 ± 30	1658 ± 26
12	75	31	325	0.09	0.00023	0.0862 ± 0.0024	2.334 ± 0.132	0.1946 ± 0.0079	1344 ± 55	1223 ± 40	1146 ± 42
13	158	47	506	0.09	0.00010	0.0872 ± 0.0010	3.269 ± 0.107	0.2683 ± 0.0040	1366 ± 23	1474 ± 25	1532 ± 20
14	245	74	769	0.10	0.00005	0.0871 ± 0.0010	3.229 ± 0.098	0.2667 ± 0.0043	1362 ± 23	1464 ± 24	1524 ± 22
15	72	12	301	0.04	0.00015	0.0861 ± 0.0030	2.390 ± 0.092	0.2016 ± 0.0054	1343 ± 66	1240 ± 27	1184 ± 29

Errors are 1σ; Pb refers to total Pb; Pb^{a} indicates the radiogenic lead portion.

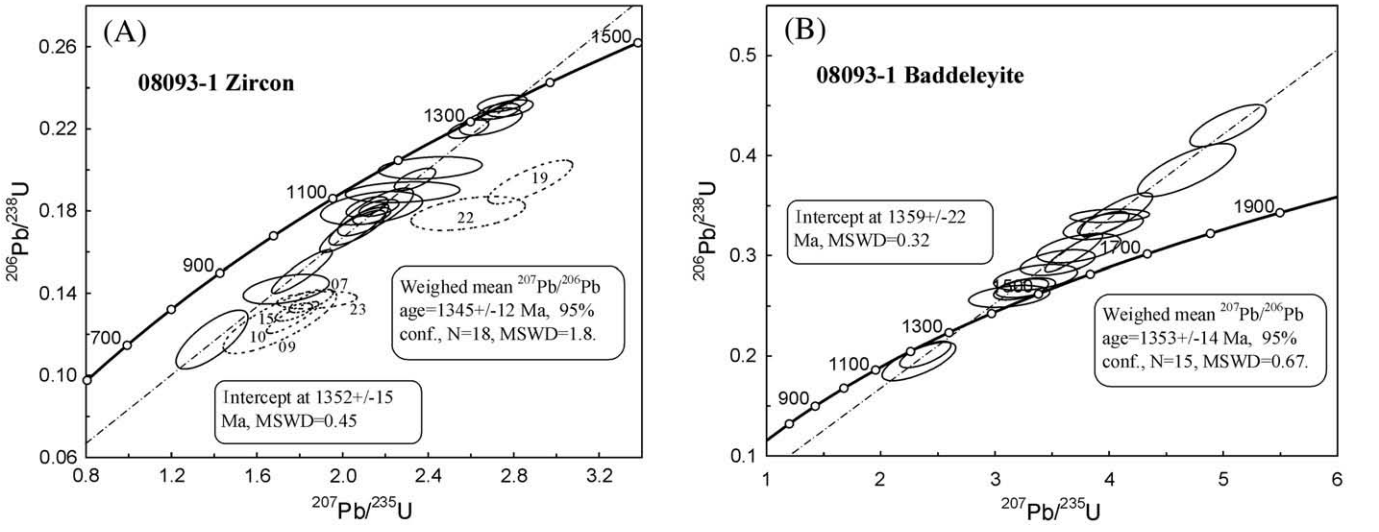


Fig. 8. U–Pb concordia diagrams for zircons (A) and baddeleyites (B) from a diabase sill sample emplaced into the Wumishan formation southwest to Chaoyang. Data-point error crosses are 2σ.

Bayan Obo-Huade-Weichang rift zone along the northern margin of the craton was formed in association with the fragmentation of the NCC from other cratonic blocks during the breakup of Columbia supercontinent (1.6–1.2 Ga), but there are no reliable isotopic ages for Mesoproterozoic magmatism or tectonism during this period. Because of the lack of reliable Meso-Neoproterozoic (1.6–1.0 Ga) thermo-tectonic records, most researchers believe that the continental rifting in the northern NCC ended prior to 1.6 Ga (e.g., Rogers and Santosh, 2002; Zhai, 2004; Lu et al., 2008), and that the NCC was not involved in the final breakup of the Columbia supercontinent during 1.35 to 1.20 Ga (e.g., Rogers and Santosh, 2002; Hou et al., 2008) nor the assembly and breakup of the Rodinia supercontinent during the late Mesoproterozoic to Neoproterozoic (e.g., Lu et al., 2008).

However, the 1.35 Ga diabase sills in the northern NCC recognized in this contribution provide important evidence for the breakup of the NCC from the Columbia supercontinent. The emplacement of these large volumes of the Mid-Mesoproterozoic (Ectasian) mafic sills into the Mesoproterozoic sedimentary rocks in the northern NCC indicates

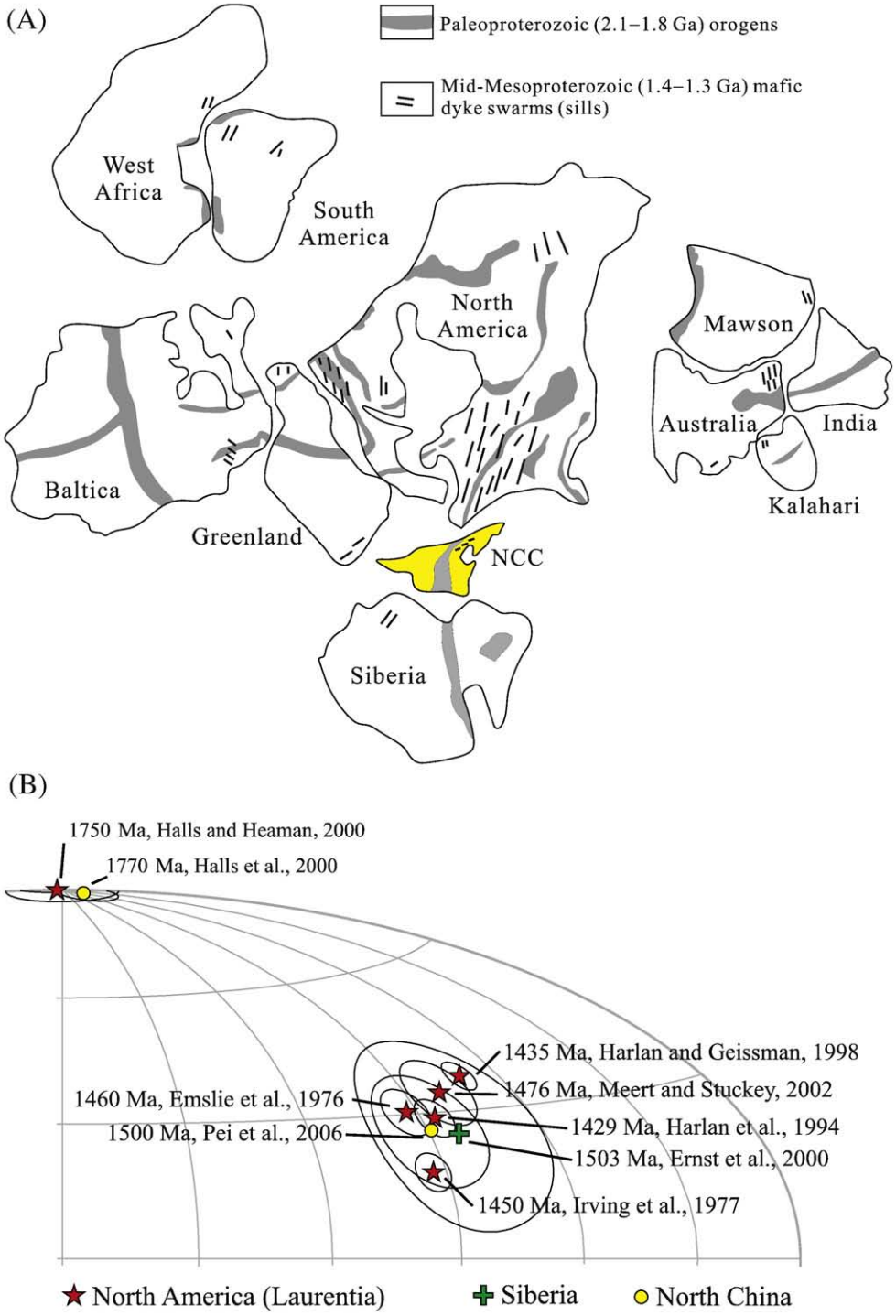


Fig. 9. Schematic cartoon (A) showing possible position of the NCC within the Columbia (Nuna) supercontinent during 1.70–1.35 Ga (modified after Rogers and Santosh, 2002; Zhao et al., 2002, 2004a). Ages of mafic dyke swarms (sills) are from Ernst et al. (2008). The position of the NCC is based on 1.35 Ga mafic sills and paleomagnetic data. The paleomagnetic evidences for connections of the NCC with Laurentia (North America) and Siberia during the Mesoproterozoic are shown in diagram (B). The position of the NCC is placed after rotating the NCC around a Eular pole at 4°N/160°E, with an angle of 53°, in which the coeval paleopoles from the NCC and Laurentia are merged.

that the northern NCC underwent strong extension during this period, suggesting that the continental rifting is still active during the Mid-Mesoproterozoic. In a global view, the Mesoproterozoic (1.6–1.2 Ga) continental rifting was recorded along both the margins and interior of Columbia (Rogers and Santosh, 2002; Zhao et al., 2002, 2004a). The ages of the mafic sills in the northern NCC are similar to those in other regions such as North America, Greenland, Siberia, Antarctica, Baltica, Africa and Australia (Ernst et al., 2008), which are regarded as being related to the final breakup of Columbia supercontinent during 1.3–1.2 Ga (Zhao et al., 2002, 2004a; Hou et al., 2008). Therefore, we suggest that the NCC was strongly involved in the breakup of Mesoproterozoic supercontinent Columbia, as evidenced by the extensive distribution of the 1.35 Ga diabase sills in the northern NCC.

Although some researchers have tried to place the NCC into the Paleo-Mesoproterozoic supercontinent (e. g., Condie, 2002; Wilde et al., 2002; Zhao et al., 2003a; Hou et al., 2008; Rogers and Santosh, 2009), the position of the NCC in the Columbia supercontinent and its connections with other cratons are not well defined because of the lack of reliable paleomagnetic and geochronological data. The NCC has been connected to Siberia, Baltica and India by Condie (2002), Wilde et al. (2002) and Zhao et al. (2003a, b), respectively. Recent paleomagnetic results on the early Mesoproterozoic Yangzhuang Formation (1.6–1.5 Ga?) in the northern NCC indicate that the NCC, Siberia, Baltica and Laurentia (North America and Greenland) might drift together or close during the Mesoproterozoic (Wu et al., 2005; Pei et al., 2006), similar to the results obtained from the late Paleoproterozoic mafic dykes (zircon U–Pb age 1769 ± 3 Ma) in the NCC (Halls et al., 2000). Comparison of the paleomagnetic data of the NCC (Halls et al., 2000; Wu et al., 2005; Pei et al., 2006) with those of the Laurentia (Emslie et al., 1976; Irving et al., 1977; Harlan et al., 1994; Harlan and Geissman, 1998; Halls and Heaman, 2000; Meert and Stuckey, 2002) and Sibiria (Ernst et al., 2000) shows that the NCC shares similar paleomagmatic poles with Laurentia and Siberia during 1.79–1.40 Ga (Fig. 9), indicating that they might drift together or close during this period. On the other hand, the 1.4–1.3 Ga mafic dyke swarms (sills) are mainly distributed in North America, Greenland, Siberia, Baltica, Antarctica and Australia (Ernst et al., 2008), we thus inferred that the NCC is a member of Nena including the Laurentia, Siberia and Baltica cratons in the Columbia supercontinent (Fig. 9). Combined with the distribution of the Mesoproterozoic (1.4–1.3 Ga) dyke swarms (sills), the 2.1–1.8 Ga orogens and the available paleomagnetic data (e.g., Emslie et al., 1976; Irving et al., 1977; Harlan et al., 1994; Harlan and Geissman, 1998; Halls et al., 2000; Halls and Heaman, 2000; Ernst et al., 2000; Meert and Stuckey, 2002; Wu et al., 2005; Pei et al., 2006), we suggested that the northern and southern margins of the NCC were likely connected to the North America (Laurentia) and Siberia cratons during the late Paleoproterozoic to Mid-Mesoproterozoic, respectively (Fig. 9).

Because the youngest ages for the Proterozoic rifting magmatism previously obtained in the NCC is around 1.6 Ga, Rogers and Santosh (2002) inferred that separation of the NCC from Baltica in the Columbia supercontinent occurred during 1.7–1.6 Ga. However, this inference is inconsistent with the recent paleomagnetic results obtained from the early Mesoproterozoic Yangzhuang Formation in the northern NCC (Wu et al., 2005; Pei et al., 2006). Because large volumes of mafic dyke swarms (sills) are important indicators for continental breakup (e.g., Holm et al., 2006; Ernst et al., 2008; Hou et al., 2008), the newly recognized 1.35 Ga diabase sills in the northern NCC indicate breakup of the NCC away from Columbia (Nuna) supercontinent during Mid-Mesoproterozoic time. Breakup of the NCC from the Columbia supercontinent during the Mid-Mesoproterozoic to late Mesoproterozoic is also supported by the paleomagnetic results on the late Mesoproterozoic sedimentary rocks (Yunmengshan Formation, Rb–Sr age ca. 1270 Ma), which display a slightly different configuration involving a ca. 90° vertical-axis rotation of the NCC with respect to Laurentia (Zhang et al., 2006; Li et al., 2008).

5.3. Constraint on the ages of the Meso-Neoproterozoic strata in the NCC

Because of the lack of index fossils within the Meso-Neoproterozoic sedimentary rocks in the NCC, the Xiamaling Formation was formerly considered as Neoproterozoic in age (e.g., Lu et al., 2008), mainly based on imprecise glauconite K–Ar or Rb–Sr ages. However, recent zircon SHRIMP U–Pb dating of K-bentonite beds within the middle segment of the Xiamaling Formation yields a narrow range of $^{207}Pb/^{206}Pb$ ages from 1366 ± 9 Ma to 1380 ± 36 Ma (Gao et al., 2007, 2008b; Su et al., 2008), indicating that it is Mesoproterozoic in age. Because the uppermost segment of the Xiamaling Formation was intruded by the diabase sills, the age for the upper boundary of the Xiamaling Formation in the Qingbaikou System is no later than 1.35 Ga. The diabase sills provide other important evidence for deposition of the Xiamaling Formation during the Mid-Mesoproterozoic instead of the Neoproterozoic. The Neoproterozoic strata are only limitedly distributed in the northern NCC, and the sedimentary gap between the Jingeryu Formation and Lower Cambrian in the northern NCC is much longer than previously regarded (e.g., Lu et al., 2008).

6. Conclusions

(1) New LA-ICP-MS U–Pb dating on zircon and baddeleyite grains from a diabase sample emplaced into the Wumishan Formation in the northern NCC yielded weighed mean $^{207}Pb/^{206}Pb$ ages of 1345 ± 12 Ma and 1353 ± 14 Ma, respectively, indicating that the diabase sills within the Mesoproterozoic sedimentary rocks in the northern NCC were emplaced during Mid-Mesoproterozoic (Ectasian) instead of the Late Paleozoic or Mesozoic as previously proposed.

(2) Existence of the large volumes of 1.35 Ga diabase sills intruded into the Mesoproterozoic marine clastic and carbonate platformal sedimentary rocks in the northern NCC indicates that the northern NCC underwent a period strong extension and continental rifting magmatism during the Mid-Mesoproterozoic (Ectasian), which is likely related to breakup of the Columbia supercontinent. The NCC is predicted to be a member of Nena including the Laurentia (North America and Greenland), Siberia and Baltica cratons in the supercontinent. The northern and southern margins of NCC were likely connected to the North America (Laurentia) and Siberia cratons during the late Paleoproterozoic to Mesoproterozoic, respectively; breaking of the NCC away from the Columbia supercontinent occurred from 1.35 Ga, which is also supported by the paleomagnetic results.

(3) The age for the upper boundary of the Xiamaling Formation in the Qingbaikou System in the northern NCC is no later than 1.35 Ga. The Xiamaling Formation was deposited during the Mid-Mesoproterozoic instead of the Neoproterozoic. The Neoproterozoic strata are only limitedly distributed in the northern NCC, and the sedimentary gap between the Jingeryu Formation and Cambrian strata in the northern NCC is much longer than previously regarded.

Acknowledgements

This research was financially supported by the National Basic Research Program of China (2006CB403501), the Institute of Geomechanics (DZLXJK200704), the Program of Excellent Young Scientists of the Ministry of Land and Resources and the "Deep Exploration Technology and Experimentation" Program of China (Sinoprobe 08-01). We thank Y.S. Liu, Z.C. Hu and X.M. Liu for the help during LA-ICP-MS U–Pb analysis. Careful and constructive reviews by Guochun Zhao, Richard E. Ernst and Richard W. Carlson (Editor) significantly improved the quality of the manuscript.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.epsl.2009.10.023.

References

Andersen, T., 2002. Correction of common lead in U–Pb analyses that do not report ^{204}Pb. Chem. Geol. 192, 59–79.

Anderson, J., Cullers, R.L., 1999. Paleo- and Mesoproterozoic granite plutonism of Colorado and Wyoming. Rocky Mt. Geol. 34, 149–164.

Bleeker, W., Ernst, R., 2006. Short-lived mantle generated magmatic events and their dyke swarms: the key unlocking Earth's paleogeographic record back to 2.6 Ga. In: Hanski, E., Mertanen, S., Rämö, T., Vuollo, J. (Eds.), Dyke Swarms: Time Markers of Crustal Evolution. Taylor and Francis/Balkema, London, pp. 3–26.

Condie, K.C., 2002. Breakup of a Paleoproterozoic supercontinent. Gondwana Res. 5, 41–43.

Dalziel, I.W.D., 1991. Pacific margins of Laurentia and East Antarctica–Australia as a conjugate rift pair: evidence and implications for an Eocambrian supercontinent. Geology 19, 598–601.

Emslie, R.F., Irving, E., Park, J.K., 1976. Further paleomagnetic results from the Michikamau intrusion, Labrador. Can. J. Earth Sci. 13, 1052–1057.

Ernst, R.E., Buchan, K.L., Hamilton, M.A., Okrugin, A.V., Tomshin, M.D., 2000. Integrated paleomagnetism and U–Pb geochronology of mafic dikes of the eastern Anabar shield region, Siberia: implications for Mesoproterozoic paleolatitude of Siberia and comparison with Laurentia. J. Geol. 108, 381–401.

Ernst, R.E., Wingate, M.T.D., Buchan, K.L., Li, Z.X., 2008. Global record of 1600–700 Ma Large Igneous Provinces (LIPs): implications for the reconstruction of the proposed Nuna (Columbia) and Rodinia supercontinents. Precambrian Res. 160, 159–178.

Gao, L.Z., Zhang, C.H., Shi, X.Y., Zhou, H.R., Wang, Z.Q., 2007. Zircon SHRIMP U–Pb dating of the tuff bed in the Xiamaling Formation of the Qingbaikouan System in North China. Geol. Bull. China 26, 249–255 (in Chinese with English abstract).

Gao, L., Zhang, C., Yin, C., Shi, X., Wang, Z., Liu, Y., Liu, P., Tang, F., Song, B., 2008a. SHRIMP zircon ages: basis for refining the chronostratigraphic classification of the Meso- and Neoproterozoic strata in North China Old Land. Acta Geoscientica Sinica 29, 366–376 (in Chinese with English abstract).

Gao, L.Z., Zhang, C.H., Shi, X.Y., Song, B., Wang, Z.Q., Liu, Y.M., 2008b. Mesoproterozoic age for Xiamaling Formation in North China Plate indicated by zircon SHRIMP dating. Chin. Sci. Bull. 53, 2665–2671.

Halls, H.C., Heaman, L.M., 2000. The paleomagnetic significance of new U–Pb age data from the Molson dyke swarm, Cauchon Lake area, Manitoba. Can. J. Earth Sci. 37, 957–966.

Halls, H.C., Li, J.H., Davis, D., Hou, G., Zhang, B.X., Qian, X.L., 2000. A precisely dated Proterozoic palaeomagnetic pole from the North China craton, and its relevance to palaeocontinental reconstruction. Geophys. J. Int. 143, 185–203.

Harlan, S.S., Geissman, J.W., 1998. Paleomagnetism of the Middle Proterozoic Electra Lake Gabbro, Needle Mountains, southwestern Colorado. J. Geophys. Res. 103, 15497–15507.

Harlan, S.S., Snee, L.W., Geissman, J.W., Brearley, A.J., 1994. Paleomagnetism of the Middle Proterozoic Laramie anorthosite complex and Sherman Granite, southern Laramie Range, Wyoming and Colorado. J. Geophys. Res. 99, 17997–18020.

Harris, L.B., Li, Z.X., 1995. Palaeomagnetic dating and tectonic significance of dolerite intrusions in the Albany Mobile Belt, Western Australia, Earth Planet. Sci. Lett. 131, 143–164.

HBGMR (Hebei Bureau of Geology and Mineral Resources), 1969. Geological map of Qinglong, Scale 1:200000 (in Chinese).

HBGMR, 2003. Geological map of Zhangbeixian (K50C003001), Scale 1:250000 (in Chinese).

Heaman, L.M., 2009. The application of U–Pb geochronology to mafic, ultramafic and alkaline rocks: an evaluation of three mineral standards. Chem. Geol. 261, 43–52.

Heaman, L.M., LeCheminant, A.N., 1993. Paragenesis and U–Pb systematics of baddeleyite (ZrO_2). Chem. Geol. 110, 95–126.

He, Y.H., Zhao, G.C., Sun, M., Wilde, S.A., 2008. Geochemistry, isotope systematics and petrogenesis of the volcanic rocks in the Zhongtiao Mountain: an alternative interpretation for the evolution of the southern margin of the North China Craton. Lithos 102, 158–178.

Hoffman, P.F., 1991. Did the breakout of Laurentia turn Gondwanaland inside-out? Science 252, 1409–1412.

Holm, P.M., Heaman, L.M., Pedersen, L.E., 2006. Baddeleyite and zircon U–Pb ages from the Kaerven area, Kangerlussuaq: implications for timing of Paleogene continental breakup in the North Atlantic. Lithos 92, 238–250.

Hou, G., Santosh, M., Qian, X., Lister, G.S., Li, J., 2008. Tectonic constraints on 1.3–1.2 Ga final breakup of Columbia supercontinent from a giant radiating dyke swarm. Gondwana Res. 14, 561–566.

IMBGMR (Inner Mongolia Bureau of Geology and Mineral Resources), 1997. Geological map of Wujiacun (K49E018013), Wuchuanxian (K49E018014), Siheyi (K49E018015) and Huanghuawopu (K49E018016), Scale 1:50000 (in Chinese).

IMBGMR, 1999. Geological map of Dongshengmiao (K48E018021), Scale 1:50000 (in Chinese).

Irving, E., Emslie, R.F., Park, J.K., 1977. Paleomagnetism of the Harp Lake Complex and associated rocks. Can. J. Earth Sci. 14, 1187–1201.

Kamo, S.L., Gower, C.F., Krogh, T.E., 1989. Birthdate for the Iapetus Ocean? A precise U–Pb zircon and baddeleyite age for the Long Range dikes, southeast Labrador. Geology 17, 602–605.

Karlstrom, K.E., Ahall, K.I., Harlan, S.S., Williams, M.L., McLelland, J., Geissman, J.W., 2001. Long-lived (1.8–1.0 Ga) convergent orogen in southern Laurentia, its extensions to Australia and Baltica, and implications for refining Rodinia. Precambrian Res. 111, 5–30.

Krogh, T.E., Corfu, F., Davis, D.W., Dunning, G.R., Heaman, L.M., Kamo, S.L., Machado, N., Greenough, J.D., Nakamura, E., 1987. Precise U–Pb isotopic ages of diabase dykes and mafic to ultramafic rocks using trace amounts of baddeleyite and zircon. In: Halls, H.C., Fahrig, W.F. (Eds.), Mafic Dyke Swarms. Geol. Assoc. Can. Special Paper, pp. 147–152.

Kröner, A., Cordani, U., 2003. African, southern Indian and South American cratons were not part of the Rodinia supercontinent: evidence from field relationships and geochronology. Tectonophysics 375, 325–352.

Kröner, A., Wilde, S.A., Zhao, G.C., O'Brien, P.J., Sun, M., Liu, D.Y., Wan, Y.S., Liu, S.W., Guo, J.H., 2006. Zircon geochronology of mafic dykes in the Hengshan Complex of northern China: evidence for late Paleoproterozoic rifting and subsequent high-pressure event in the North China Craton. Precambrian Res. 146, 45–67.

LBGMR (Liaoning Bureau of Geology and Mineral Resources), 1965. Geological map of Lingyuan, Scale 1:200000 (in Chinese).

LBGMR, 1967. Geological map of Chaoyang, Scale 1:200000 (in Chinese).

LBGMR, 1969, Geological map of Jinxi, Scale 1:200000 (in Chinese).

Li, W.P., Li, X.H., 2005. Geochemical characteristics of the Late Paleozoic diabase dyke swarms of Changmaohezi from western Liaoning, Northeast China. Earth Sci. 30, 761–770 (in Chinese with English abstract).

Li, H.K., Li, H.M., Lu, S.N., 1995. Grain zircon U–Pb age for volcanic rocks from Tuanshanzi Formation of Changcheng System and their geological implication. Geochimica 24, 43–48 (in Chinese with English abstract).

Li, H.M., Li, H.K., Chen, Z.H., Xiang, Z.Q., Lu, S.N., Zhou, H.Y., Song, B., 2007. Comparison of three methods for baddeleyite U–Pb isotopic dating of basic rocks. Geol. Bull. China 26, 128–135 (in Chinese with English abstract).

Li, Z.X., Bogdanova, S.V., Collins, A.S., Davidson, A., De Waele, B., Ernst, R.E., Fitzsimons, I.C.W., Fuck, R.A., Gladkochub, D.P., Jacobs, J., Karlstrom, K.E., Lu, S., Natapov, L.M., Pease, V., Pisarevsky, S.A., Thrane, K., Vernikovsky, V., 2008. Assembly, configuration, and break-up history of Rodinia: a synthesis. Precambrian Res. 160, 179–210.

Liu Y.S., Gao S., Hu Z.C., Gao C.G., Zong K.Q., Wang D.B., submitted for publication. Continental and oceanic crust recycling-induced melt–peridotite interactions in the Trans-North China Orogen: U–Pb dating, Hf isotopes and trace elements in zircons of mantle xenoliths. J. Petrol.

Lu, S.N., Li, H.M., 1991. Zircon age of Dahongyu Formation. Bull. Chin. Acad. Geol. Sci. 22, 137–145 (in Chinese).

Lu, S.N., Yang, C.L., Li, H.K., Chen, Z.H., 2002. North China continent and Columbia supercontinent. Earth Sci. Front. 9, 225–233 (in Chinese with English abstract).

Lu, S.N., Zhao, G.C., Wang, H.C., Hao, G.J., 2008. Precambrian metamorphic basement and sedimentary cover of the North China Craton: a review. Precambrian Res. 160, 77–93.

Ludwig, K.R., 2003. User's manual for Isoplot 3.00. A geochronological Toolkit for Microsoft Excel. Berkeley Geochronology Center, Special Publication No. 4a, Berkeley, California. 70 pp.

Meert, J.G., 2002. Paleomagnetic evidence for a Paleo-Mesoproterozoic supercontinent Columbia. Gondwana Res. 5, 207–215.

Meert, J.G., Stuckey, W., 2002. Revisiting the paleomagnetism of the 1.476 Ga St. Francois Mountains igneous province, Missouri. Tectonics 21, 1007. doi:10.1029/2000TC001265.

Moores, E.M., 1991. Southwest U.S.–East Antarctic (SWEAT) connection: a hypothesis. Geology 19, 425–428.

Murphy, J.B., Nance, R.D., Gutiérrez-Alonso, G., Keppie, J.D., 2009. Supercontinent reconstruction from recognition of leading continental edges. Geology 37, 595–598.

Pei, J., Yang, Z., Zhao, Y., 2006. A Mesoproterozoic paleomagnetic pole from the Yangzhuang Formation, North China and its tectonics implications. Precambrian Res. 151, 1–13.

Peng, P., Zhai, M.G., Zhang, H.F., Guo, J.G., 2005. Geochronological constraints on Paleoproterozoic evolution of the North China craton: SHRIMP zircon ages of different types of mafic dykes. Int. Geol. Rev. 47, 492–508.

Peng, P., Zhai, M.G., Ernst, R.E., Guo, J.H., Liu, F., Hu, B., 2008. A 1.78 Ga large igneous province in the North China craton: the Xiong'er Volcanic Province and the North China dyke swarm. Lithos 101, 260–280.

Rogers, J.J.W., Santosh, M., 2002. Configuration of Columbia, a Mesoproterozoic supercontinent. Gondwana Res. 5, 5–22.

Rogers, J.J.W., Santosh, M., 2003. Supercontinents in earth history. Gondwana Res. 6, 357–368.

Rogers, J.J.W., Santosh, M., 2009. Tectonics and surface effects of the supercontinent Columbia. Gondwana Res. 15, 373–380.

Shao, J.A., Zhang, L.Q., Li, D.M., 2002. Three Proterozoic extensional events in North China Craton. Acta Petrol. Sinica 18, 152–160 (in Chinese with English abstract).

Söderlund, U., Johansson, L., 2002. A simple way to extract baddeleyite (ZrO_2). Geochem. Geophys. Geosyst. 3, 1014. doi:10.1029/2001GC000212.

Su, W., Zhang, S., Huff, W.D., Li, H., Ettensohn, F.R., Chen, X., Yang, H., Han, Y., Song, B., Santosh, M., 2008. SHRIMP U–Pb ages of K-bentonite beds in the Xiamaling Formation: Implications for revised subdivision of the Meso- to Neoproterozoic history of the North China Craton. Gondwana Res. 14, 543–553.

Wang, Y.J., Fan, W.M., Zhang, Y.H., Guo, F., Zhang, H.F., Peng, T.P., 2004. Geochemical, ^{40}Ar/^{39}Ar geochronological and Sr–Nd isotopic constraints on the origin of Paleoproterozoic mafic dikes from the southern Taihang Mountains and implications for the ca. 1800 Ma event of the North China Craton. Precambrian Res. 135, 55–77.

Wilde, S.A., Zhao, G.C., Sun, M., 2002. Development of the North China Craton during the Late Archaean and its final amalgamation at 1.8 Ga; some speculations on its

position within a global Palaeoproterozoic supercontinent. Gondwana Res. 5, 85–94.
Windley, B.F., 1995. The Evolving Continents, Third Edition. John Wiley and Son Ltd, Chichester. 526 pp.
Wingate, M.T.D., 2001. SHRIMP baddeleyite and zircon ages for an Umkondo dolerite sill, Nyanga Mountains, Eastern Zimbabwe. South Afr. J. Geol. 104, 13–22.
Wingate, M.T.D., Compston, W., 2000. Crystal orientation effects during ion microprobe U–Pb analysis of baddeleyite. Chem. Geol. 168, 75–97.
Wingate, M.T.D., Campbell, I.H., Compston, W., Gibson, G.M., 1998. Ion microprobe U–Pb ages for Neoproterozoic basaltic magmatism in south-central Australia and implications for the breakup of Rodinia. Precambrian Res. 87, 135–159.
Wu, H.C., Zhang, S.H., Li, Z.X., Li, H.Y., Dong, J., 2005. New paleomagnetic results from the Yangzhuang Formation of the Jixian System, North China, and tectonic implications. Chin. Sci. Bull. 50, 1483–1489.
Xia, X.P., Sun, M., Zhao, G.C., Wu, F.Y., Xu, P., Zhang, J.H., Luo, Y., 2006. U–Pb and Hf isotopic study of detrital zircons from the Wulashan khondalites: constraints on the evolution of the Ordos Terrane, Western Block of the North China Craton. Earth Planet. Sci. Lett. 241, 581–593.
Yale, L.B., Carpenter, S.J., 1998. Large igneous provinces and giant dike swarms: proxies for supercontinent cyclicity and mantle convection. Earth Planet. Sci. Lett. 163, 109–122.
Yang, J.H., Wu, F.Y., Liu, X.M., Xie, L.W., 2005. Zircon U–Pb ages and Hf isotopes and their geological significance of the Miyun rapakivi granites from Beijing, China. Acta Petrol. Sinica 21, 1633–1644 (in Chinese with English abstract).
Yuan, H.L., Gao, S., Liu, X.M., Li, H.M., Günther, D., Wu, F.Y., 2004. Accurate U–Pb age and trace element determinations of zircon by laser ablation inductively coupled plasma mass spectrometry. Geostand. Geoanal. Res. 28, 353–370.
Zhai, M.G., Bian, A.G., Zhao, T.P., 2000. The amalgamation of the supercontinent of North China craton at the end of the Neoarchaean, and its break-up during the late Palaeoproterozoic and Mesoproterozoic. Sci. China (Ser. D) 43 (Suppl.), 219–232.
Zhai, M.G., 2004. Precambrian tectonic evolution of the North China Craton. In: Malpas, J., Fletcher, C.J.N., Ali, J.R., Aitchison, J.C. (Eds.), Aspects of the tectonic evolution of China: Geological Society Special Publications, vol. 226, pp. 57–72.
Zhang, S.H., Li, Z.X., Wu, H., 2006. New Precambrian palaeomagnetic constraints on the position of the North China Block in Rodinia. Precambrian Res. 144, 213–238.
Zhang, S.H., Liu, S.W., Zhao, Y., Yang, J.H., Song, B., Liu, X.M., 2007a. The 1.75–1.68 Ga anorthosite-mangerite-alkali granitoid-rapakivi granite suite from the northern North China Craton: magmatism related to a Paleoproterozoic orogen. Precambrian Res. 155, 287–312.
Zhang, C., Liu, S.W., Han, B.F., He, G.Q., Huang, B.L., 2007b. SHRIMP U–Pb dating of Dashigou biotite–K–felspar granites in Shangdu, Inner Mongolia, and its significance. Acta Petrol. Sinica 23, 591–596 (in Chinese with English abstract).
Zhang, C., Han, B.F., Liu, S.W., Ji, J.Q., Zhao, L., Zhang, L., 2009a. SHRIMP U–Pb dating of biotite granites in Daqingshan, Inner Mongolia, and its significance. Acta Petrol. Sinica 25, 561–567 (in Chinese with English abstract).
Zhang, S.H., Zhao, Y., Song, B., Hu, J.M., Liu, S.W., Yang, Y.H., Chen, F.K., Liu, X.M., Liu, J., 2009b. Contrasting Late Carboniferous and Late Permian–Middle Triassic intrusive suites from the northern margin of the North China craton: geochronology, petrogenesis and tectonic implications. Geol. Soc. Am. Bull. 121, 181–200.
Zhao, G.C., Cawood, P.A., Wilde, S.A., Sun, M., Lu, L., 2000. Metamorphism of basement rocks in the Central Zone of the North China craton: implications for Palaeoproterozoic tectonic evolution. Precambrian Res. 103, 55–88.
Zhao, G.C., Wilde, S.A., Cawood, P.A., Sun, M., 2001. Archean blocks and their boundaries in the North China Craton: lithological, geochemical, structural and P–T path constraints and tectonic evolution. Precambrian Res. 107, 45–73.
Zhao, G.C., Cawood, P.A., Wilde, S.A., Sun, M., 2002. A review of the global 2.1–1.8 Ga orogens: implications for a pre-Rodinian supercontinent. Earth-Sci. Rev. 59, 125–162.
Zhao, G.C., Sun, M., Wilde, S.A., Li, S.Z., 2003a. Assembly, accretion and breakup of the Paleo-Mesoproterozoic Columbia Supercontinent: records in the North China Craton. Gondwana Res. 6, 417–434.
Zhao, G.C., Sun, M., Wilde, S.A., 2003b. Correlations between the Eastern Block of the North China Craton and the South Indian Block of the Indian Shield: An Archean to Paleoproterozoic link. Precambrian Res. 122, 201–233.
Zhao, G.C., Sun, M., Wilde, S.A., Li, S., 2004a. A Paleo-Mesoproterozoic supercontinent: assembly, growth and breakup. Earth-Sci. Rev. 67, 91–123.
Zhao, T.P., Chen, F.K., Zhai, M.G., Xia, B., 2004b. Single zircon U–Pb ages and their geological significance of the Damiao anorthosite complex, Heibei Province, China. Acta Petrol. Sinica 20, 685–690 (in Chinese with English abstract).
Zhao, Y., Zhang, S.H., Xu, G., Yang, Z.Y., Hu, J.M., 2004c. The Jurassic major tectonic events of the Yanshanian intraplate deformation belt. Geol. Bull. China 23, 22–31 (in Chinese with English abstract).
Zhao, G.C., Sun, M., Wilde, S.A., Li, S.Z., 2005. Late Archean to Paleoproterozoic evolution of the North China Craton: key issues revisited. Precambrian Res. 136, 177–202.
Zhao, G.C., Sun, M., Wilde, S.A., Li, S.Z., Zhang, J., 2006. Some key issues in reconstructions of Proterozoic supercontinents. J. Asian Earth Sci. 28, 3–19.
Zhao, G.C., He, Y.H., Sun, M., 2009. The Xiong'er volcanic belt at the southern margin of the North China Craton: Petrographic and geochemical evidence for its outboard position in the Paleo-Mesoproterozoic Columbia Supercontinent. Gondwana Res. 16, 170–181.
Zhu, W., Zhang, Z., Shu, L., Lu, H., Su, J., Yang, W., 2008. SHRIMP U–Pb zircon geochronology of Neoproterozoic Korla mafic dykes in the northern Tarim Block, NW China: Implications for the long-lasting breakup process of Rodinia. J. Geol. Soc. London 165, 887–890.